

Auteur: Zaky Souai
Studierichting: Informatica
Oefeningenreeks 4A nr. 50.6

$$\begin{aligned} & \int \sin x \cos 3x dx \\ &= \int \frac{1}{2} (\sin(x + 3x) + \sin(x - 3x)) dx \\ &= \int \frac{1}{2} (\sin 4x + \sin(-2x)) dx \\ &= \int \frac{1}{2} (\sin 4x - \sin 2x) dx \\ &= \frac{1}{2} \int (\sin 4x - \sin 2x) dx \\ &= \frac{1}{2} \left(-\frac{1}{4} \cos 4x + \frac{1}{2} \cos 2x \right) + C \\ &= -\frac{1}{8} \cos 4x + \frac{1}{4} \cos 2x + C \end{aligned}$$

References